Prerna Kumari Sharma

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Career Objective

Ambitious MCA student focusing on AI, Machine Learning and modern full-stack development. Experienced in optimizing and deploying quantized deep learning models on resource-constrained devices, as well as developing robust backend services. Motivated to solve real-world engineering challenges in collaborative, high-impact environments.

Projects

Agentic Literature Review Assistant https://github.com/prerna12github/agentic-literature-review

- Built a human-in-the-loop multi-agent system (LangGraph) that automates end to end literature reviews with human review checkpoints.
- Implemented zero-cost contradiction detection using local MiniLM embeddings for claim clustering and Gemini for conflict analysis.
- Enforced strict citation provenance, deterministically tagging claims with chunk/page metadata and anti-hallucination source verification.
- Tech: Python, LangGraph, Google Gemini API, Sentence-Transformers

Accident Detection https://github.com/prerna12github/accident-detection-using-cnn

- Built a real-time CCTV accident detection pipeline using a MobileNetV2 model optimized for edge hardware.
- Applied post-training quantization and OpenCV frame preprocessing to maximize inference throughput and reduce latency.
- Tech: Python, TensorFlow, MobileNetV2, OpenCV

URL Shortener https://github.com/prerna12github/u-short

- Developed a low-latency URL shortening service featuring custom aliases and automated 30-day link expiry via pg_cron.
- Integrated Valkey (Redis) caching with PostgreSQL persistence to minimize database load and accelerate redirections.
- Tech: FastAPI, Python, PostgreSQL, Valkey / Redis, pg_cron

Education

Kalyani Government Engineering College — Kalyani, Nadia, West Bengal
Master of Computer Applications (MCA) — Pursuing — *2025 — Present*

Kalyani Mahavidyalaya (University of Kalyani) — Kalyani, Nadia, West Bengal
B.Sc. (Hons) in Computer Science — CGPA: 7.96 — *2022 — 2025*

Skills

- **Languages**: JavaScript, TypeScript, Python, Java
- **Frontend**: HTML, CSS, React, Tailwind CSS, Next.js
- **Backend**: Node.js, FastAPI, REST APIs
- **Databases**: PostgreSQL, MySQL, SQLite, Supabase, Redis
- **Tools & ML**: Git, GitHub, OpenCV, TensorFlow, Postman

Achievements

- **WBJECA Rank 62**: Secured All-Bengal Rank 62 in the West Bengal Joint Entrance Examination for Computer Applications (WBJECA).